

Food Demand Forecasting

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Machine Learning 2025/26

RESEARCH QUESTION

**"Does clustering fulfillment centers
by operational profile reduce demand
prediction error and procurement waste?"**

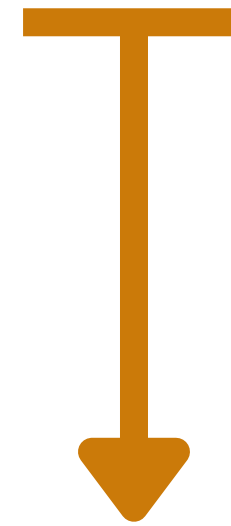
RESEARCH QUESTION

CORE HYPOTHESIS

Centers sharing similar operational profiles also share similar demand patterns
=> model trained on separate centers will learn those localized patterns more accurately

CORE UNCERTAINTY

Sufficiently powerful models could be capable of learning group-specific patterns implicitly without explicit pre-segmentation



COMPARISON

LVL A) SEGMENTED APPROACH

- **FRAMEWORK:** Dataset physically partitioned into 3 distinct, independent sub-matrices.
- **NO POOLING:** Separate models trained in complete isolation for each operational cluster.
- **CORE HYPOTHESIS:** Eliminates cross-group variance and noise, forcing the model's internal capacity to specialize 100% on localized demand dynamics.

LVL B) GLOBAL APPROACH

- **FRAMEWORK:** Single model trained on the entire unified dataset (456,548 rows).
- **INFORMED POOLING:** Integrates cluster_id as an explicit categorical feature instead of training blindly.
- **CORE HYPOTHESIS:** Maximizes statistical power and global trends, allowing the trees to isolate group-specific patterns via feature splits without losing data volume.

COMPLETE PIPELINE

PHASE A: Data preparation and clustering

01

Data Loading & Initial EDA

02

Cleaning and preprocessing

03

Multi-variate explorative analysis

04

Feature engineering pipeline

05

Unsupervised clustering

06

Stratified subsampling

PHASE B: Predictive ML Modeling

01

Metric Setup (RMSLE) & Target Splitting

02

Naive Temporal Baselines

03

Parametric Linear Baselines

04

LVL A: Global Non-Linear Models (HGBR)

05

LVL B: Cluster-Segmented Modeling

06

Robustness Probes and Business Cost

DATASETS DESCRIPTION

train.csv - 456,548 rows

Column Name	Type	Range / Values	Business/ML Role
id	int	1.0 MLN → 1.5 MLN	Unique identifier (dropped after load)
week	int	1 → 145	Chronological week timeline (~2.8 years)
center_id	int	10 → 186	Join key to fulfillment metadata
meal_id	int	1,062 → 2,956	Join key to meal metadata
checkout_price	float	2.97 → 866.27	Final price paid; used as Revenue Proxy
base_price	float	55.35 → 866.27	Undiscounted standard price; Cost Proxy
emailer_for_promotion	binary	0 or 1	Sparse indicator for email campaigns
homepage_featured	binary	0 or 1	High-visibility app/web highlight banner
num_orders	int	13 → 24,299	Target Variable (highly right-skewed)

fulfilment_center_info.csv - 77 rows

Column Name	Type	Range / Values	Business/ML Role
center_id	int	10 → 186	Primary Key for spatial operations
city_code	int	456 → 713	Anonymized geographic city marker
region_code	int	23 → 93	Anonymized regional cluster identifier
center_type	cat	TYPE_A, B, C	Categorical classification of facility ops
op_area	float	0.90 → 7.00	Physical center footprint size metrics

meal_info.csv - 51 rows

Column Name	Type	Unique Values	Business/ML Role
meal_id	int	51 unique IDs	Primary Key for catalog properties
category	cat	14 distinct types	Dish style (Beverages, Pizza, Pasta, etc.)
cuisine	cat	4 traditions	Thai, Indian, Italian (most ordered), Continental

the 3 datasets have been merged together into one, using as keys meal_id and center_id

DATASETS DESCRIPTION

ANONYMIZATION

all identifiers (center_id, meal_id, city_code, region_code, center_type) are anonymous

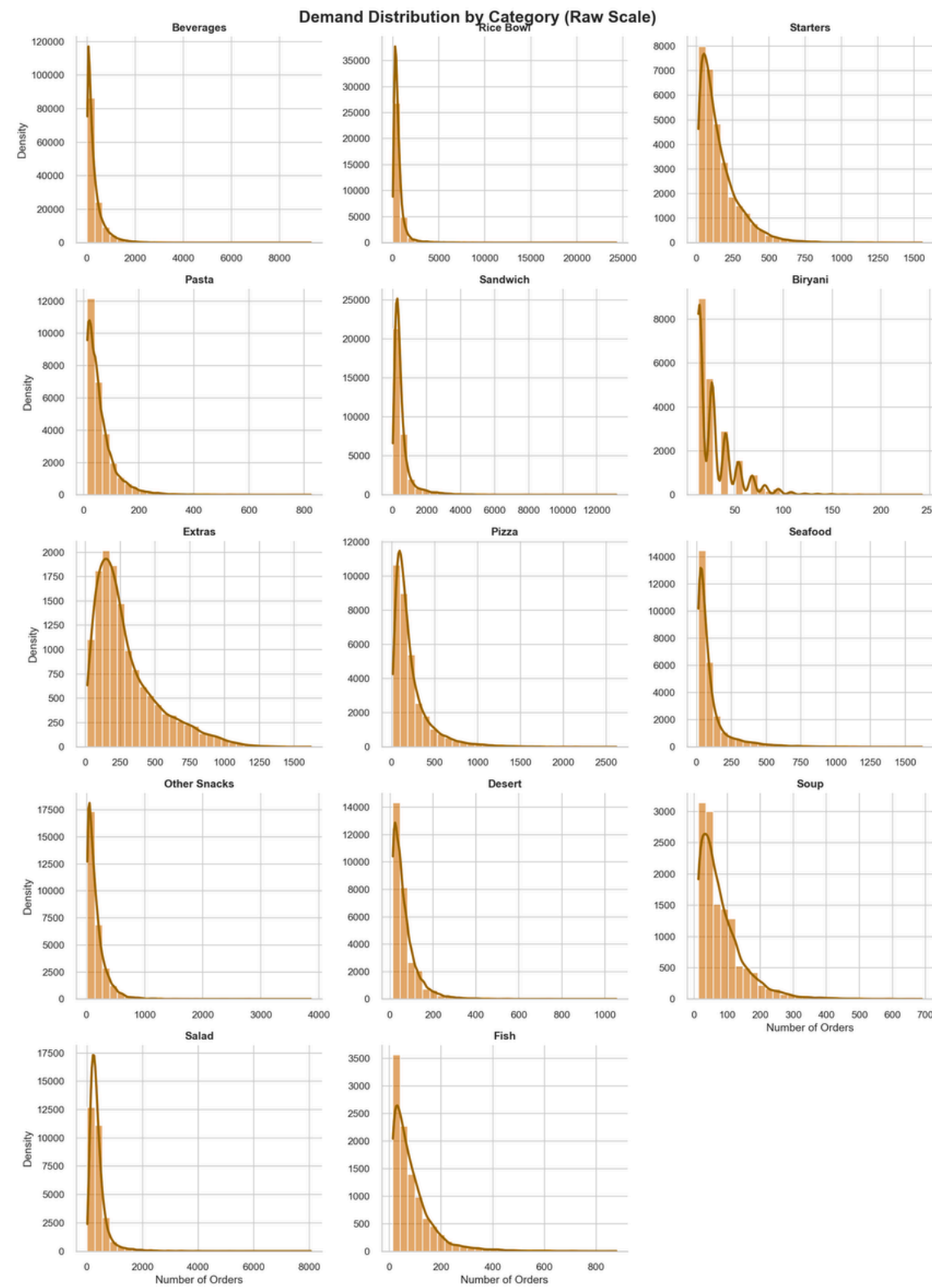
NO TIMESTAMPS

only relative week numbers are available, can't anchor to real calendar events

NO ZERO ORDER ROWS

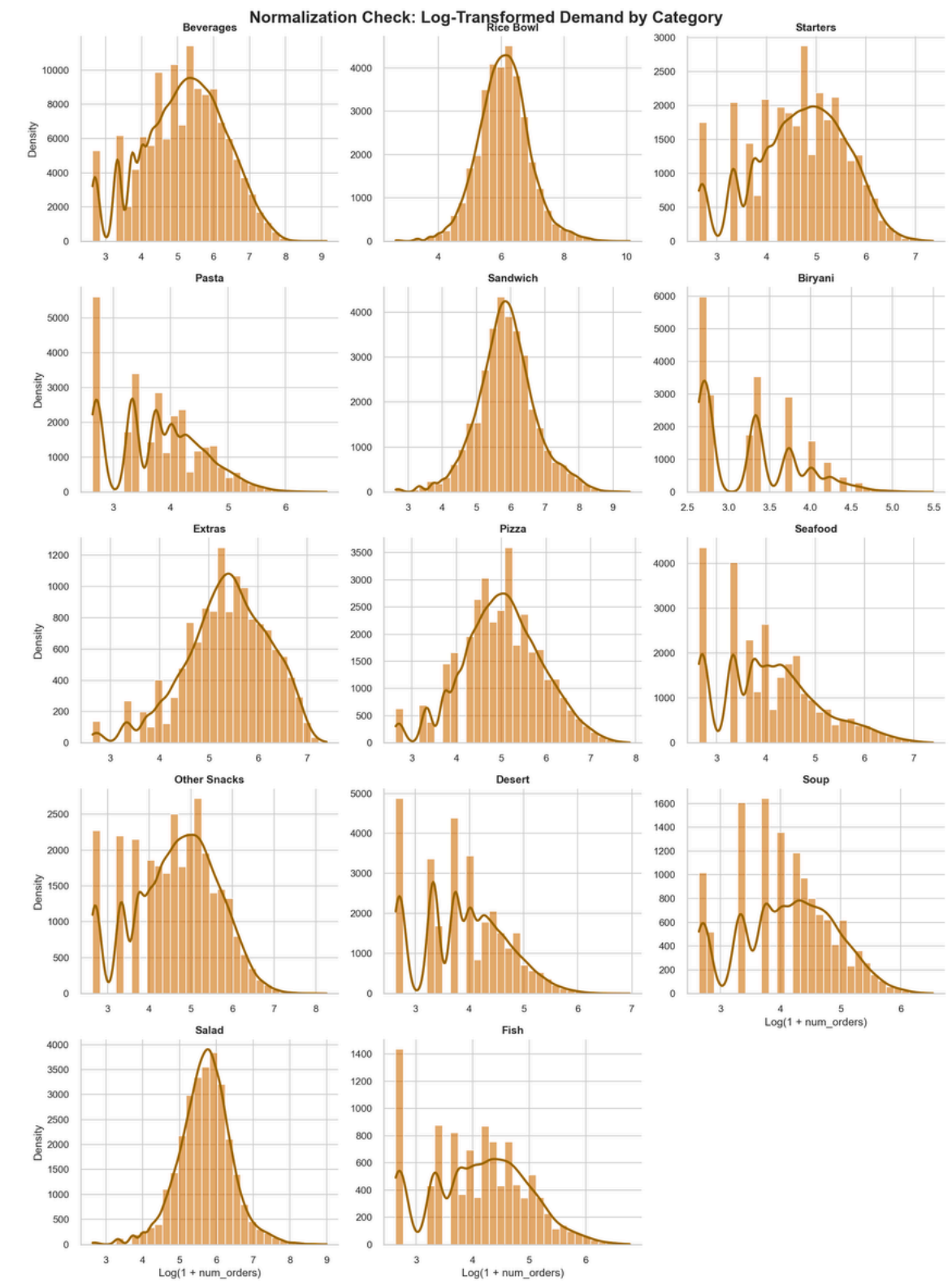
weeks with 0 orders for a center-meal are absent from the training data, rather than recorded as 0

DATA TRANSFORMATIONS



LOGARITHMIC TRANSFORMATION

of the target variable
“num_orders”



FEATURE ENGINEERING

Converting raw relational schemas into deliberate mathematical representations of market dynamics.
Added into the main training dataset

CROSS SECTIONAL FEATURES

discount_amount

Absolute raw price reduction. Strongest baseline drivers

discount_pct

Absolute raw price reduction. Strongest baseline drivers

discount_flag

Absolute raw price reduction. Strongest baseline drivers

combined_promo

Absolute raw price reduction. Strongest baseline drivers

TEMPORAL FEATURES

lag_4, lag_8

medium-term demand cycles per center-meal pair

rolling_mean_12

Long-window macro trajectory tracker

rolling_std_4

Short-windows local transaction volatility & noise index

week_of_year

Cuclical calendar index capturing annual seasonality



Leakage Prevention via **Temporal Offsets**
to prevent lookahead bias, by doing
.shift(1)

CLUSTERING

SCALE PREDICTABILITY

- **The Metric:** Combines **average weekly volume** (mean_demand) with its **stability over time**.
- **The Safeguard:** Uses the **Coefficient of Variation** instead of raw volatility.
- **The Goal:** **Neutralizes the "size effect"**—preventing massive regional hubs from dominating the geometric distance and drowning out small fulfillment centers.

PROMO SENSITIVITY

- **The Metric:** Calculates a **demand multiplier** showing **how much business surges when a marketing campaign hits**.
- **The Data Challenge:** Quiet centers with zero historical promotion exposure create missing data gaps (NaN).
- **The Fix:** **Missing metrics are neutralized to 1.0** (the baseline for "no change"), preventing the algorithm from misclassifying untested sites as failures.

PRICE ELASTICITY

- **The Metric:** Uses the **correlation between price and demand** to track how **sensitive** customers are to cost changes.
- **The Logic:** A **strong negative correlation flags price-sensitive markets**; a flat correlation flags inelastic areas where customers buy regardless of price.
- **The Goal:** **Separates regions where discounts drive massive volume** from strict, inelastic constraints where price changes have zero impact.

PHYSICAL CONSTRAINTS

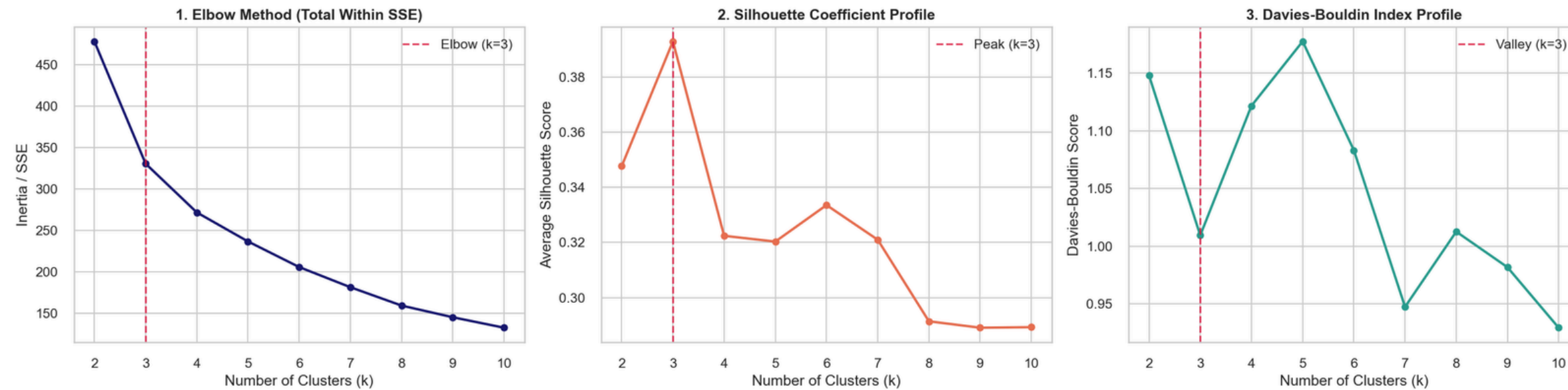
- **The Metric:** Maps the concrete reality of each facility using its **physical size (op_area)** and **facility type**.
- **The Logic:** customer behaviors (like buying spikes) are limited by physical infrastructure. A small urban outpost cannot process the volume of a massive regional warehouse.
- **The Goal:** **ensuring clusters respect the physical capacity and structural limits** of each site.

FINAL UNIFIED FEATURE ARRAY MATRIX 77*8

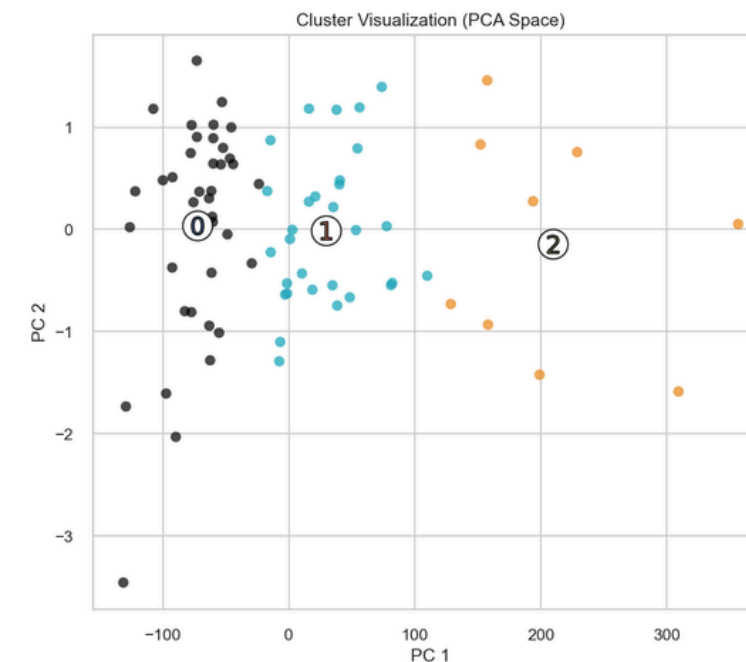
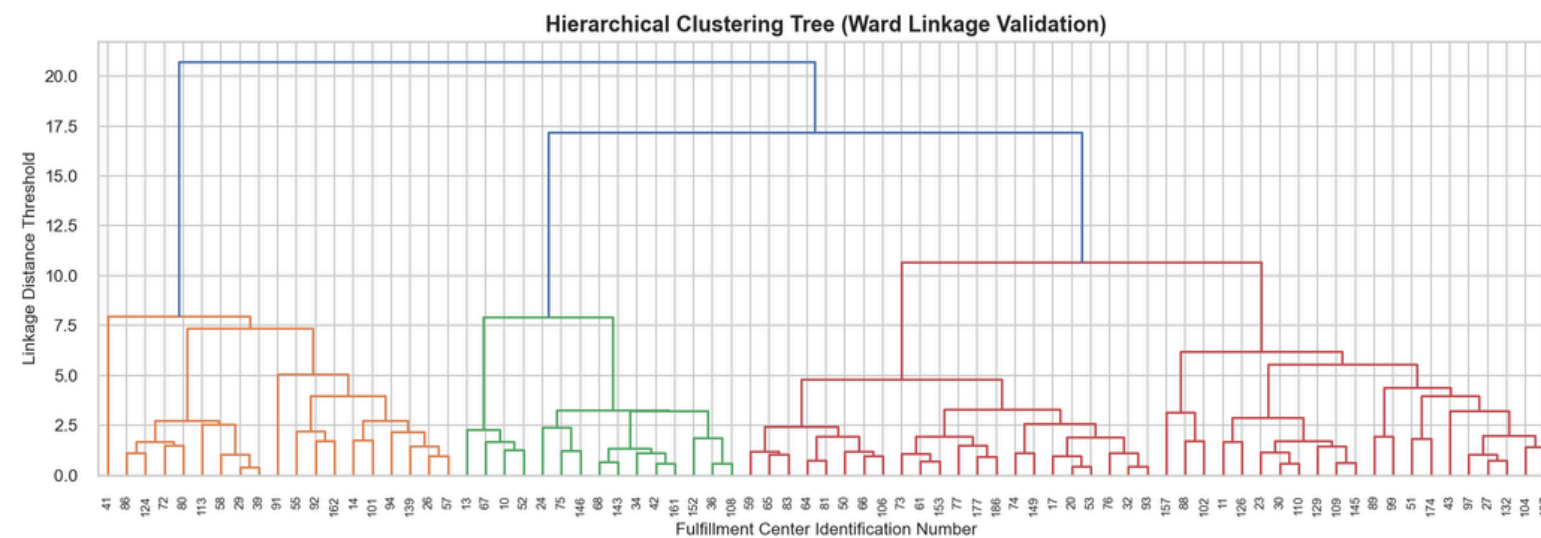
- mean_demand
- hoemapge_uplift
- price_elasticity_proxy
- center_type
- coefficient_of_variant
- emailer_uplift
- op_area
- city_region_markers

CLUSTERING

Unsupervised Internal Validation Indices for Optimal k Selection



- > **SSE**: within-cluster compactness
- > **Silhouette-score**: measures cluster assignment confidence. Higher is better
- > **Davies-Bouldin Index**: ratio of within-cluster distances to between-cluster distances. Lower is better



k = 3

BUSINESS INTERPRETATION

Cluster ID	Mean Weekly Demand	Volatility (CV)	Email Sensitivity	Homepage Sensitivity	Price Elasticity	Operational Area	Dominant Center Type	Network Count
Cluster 0	180.492	1.332	2.720	2.827	-278	3.400	TYPE_A	38 Centers
Cluster 1	282.605	1.403	3.084	2.689	-313	4.277	TYPE_A	30 Centers
Cluster 2	462.479	1.302	2.846	2.545	-377	5.489	TYPE_B	9 Centers

CLUSTER 0 n= 38 STANDARD BASELINE

- Small-to-moderate footprint facilities
- lower average demand.
- consistent baseline volatility
- predictable, standard marketing reactivity

CLUSTER 1 n= 30 PROMO-SENSITIVE

- Mid-sized hubs
- highest overall demand volatility.
- intensely reactive to marketing initiatives, with an exceptional 3.08x demand surge when targeted by email promotions

CLUSTER 2 n= 9 HIGH-VOLUME HUBS

- Massive regional fulfillment complexes
- highest volume operations
- low, stable volatility margins
- strong macroeconomic exposure via elevated price elasticity constraints (-0.377)

SUBSAMPLING

GOAL: reduce the number of rows in the dataset to speed up training

rows = 300000

STRATIFIED SUBSAMPLING

- **cluster id**

stratification keys

- **cuisine type**



prevents data scarcity for some categories



controls that cuisine buckets maintain critical mass variance and representation

CHRONOLOGICAL SEQUENCE

used for main pipeline

2 SUBSAMPLES

- real proportions

based on cluster_id

- **forced balanced**



preserves the real data-generating process



used as a robustness check against cluster-size imbalance

Predictive Modelling

PHASE B — PREDICTIVE MODELLING

“Does segmenting fulfillment centers by operational profile reduce demand prediction error and procurement waste?”

LEVEL A — GLOBAL

One model on the full subsample;
cluster_id given as a feature.
Maximises statistical power.

LEVEL B — SEGMENTED

Data physically split by cluster;
one HGBR trained in isolation per
operational archetype.

LINEAR NULL

Linear & Ridge as the empirical
floor. Every model is judged on
chronological-holdout RMSLE.

HOW WE MEASURE & VALIDATE

1

STATISTICAL METRIC

$$\text{RMSLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(1 + \hat{y}_i) - \log(1 + y_i))^2}$$

Penalises proportional error.
A 20% miss costs the same on low- and high-volume meals, mirroring procurement waste.

2

CHRONOLOGICAL SPLIT

Train = weeks 1–135
Validation = weeks 136–145

Strictly forward-looking.
A random split would leak future weeks into training and inflate accuracy.

3

OPERATIONAL METRICS

$$\text{Waste} = \sum_{i=1}^n \max(0, \hat{y}_i - y_i) \cdot \text{checkout_price}_i$$

$$\text{Lost Sales} = \sum_{i=1}^n \max(0, y_i - \hat{y}_i) \cdot \text{base_price}_i$$

Reveals the asymmetry RMSLE hides:
all good models are shortage-biased,
i.e. lose more revenue from stockouts
than they waste in excess

NAÏVE BASELINES: THE NULL FLOOR

Two training-free rules set the floor every learned model must beat

LAST-WEEK · predict = log(orders $t-1$)

0.7072

A single recent week is a noisy predictor.

ROLLING MEAN · 12-week average

0.6284

≈ 11% better — demand follows its trend.

THE BENCHMARK TO BEAT: 0.6284

Any model — linear, ensemble, or segmented — must improve on the rolling-mean rule to justify its complexity over something a procurement analyst could compute by hand. The gap between the two rules already shows demand at the center-meal level is driven by medium-term behaviour, not week-to-week noise.

LINEAR & RIDGE REGRESSION

The additive null hypothesis — does modelling price & promotion beat the floor?

Linear Regression

0.5270

Ridge (CV-tuned $\alpha = 1000$)

0.5271

≈ 16% GAIN OVER THE NAÏVE FLOOR

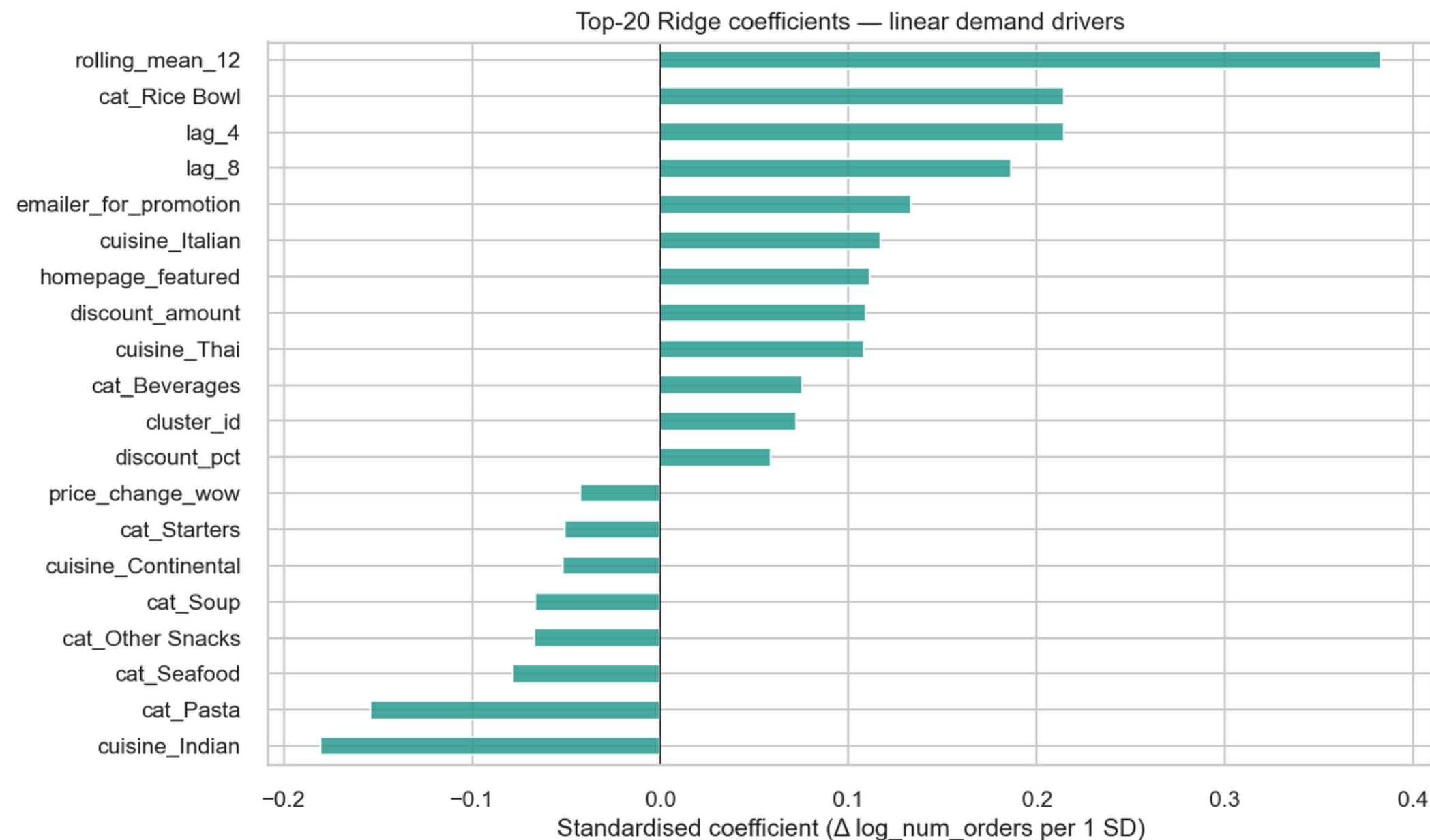
Modelling price & promotion adds real, measurable signal.

WHAT RIDGE TELLS US

- **Strong regularisation ($\alpha = 1000$)** selected at the top of the grid — a signal of heavy collinearity among price & temporal features.
- **Ridge $\approx$ Linear (-0.0002)** means regularisation buys coefficient stability, not accuracy. We read drivers off Ridge for that reason.
- **Additive structure is the ceiling** and each feature acts independently — real demand has interactions linear models cannot represent.
- **Motivates non-linear ensembles** to capture price $\times$ promotion $\times$ temporal interactions the linear model misses.

TOP-20 RIDGE COEFFICIENTS

Standardised coefficients — the strongest linear demand drivers



READING THE WEIGHTS

- **rolling_mean_12 dominates ($\beta \approx 0.40$)** - recent demand history carries most of the linear signal — validating the temporal feature engineering.
- **lag_4 & lag_8 follow** - medium-term demand cycles are the next strongest predictors.
- **Menu effects appear next** - cat_Rice Bowl, cuisine_Italian/Indian shift demand by category.
- **city_code, discount_flag ≈ 0** - under strong shrinkage Ridge ignores predictors with no marginal linear value.

NON-LINEAR ENSEMBLES (LEVEL A)

Random Forest vs HGBR — two architectures across the bias–variance trade-off

RANDOM FOREST

Parallel bagged trees aggregated by averaging — primarily variance reduction.

0.5038 RMSLE

HGBR

Sequential trees correcting residuals — primarily bias reduction. Histogram bins keep it fast.

0.4984 RMSLE

THEY CONVERGE

Only 0.006 RMSLE apart — inside the noise floor. Two opposite mechanisms reach the same accuracy: the pipeline is nearing the dataset's **predictability ceiling**.

TUNING (RF, TimeSeriesSplit CV)

10 sampled configs · carried forward as Level A

CV RMSLE range

0.519–0.547

Best config

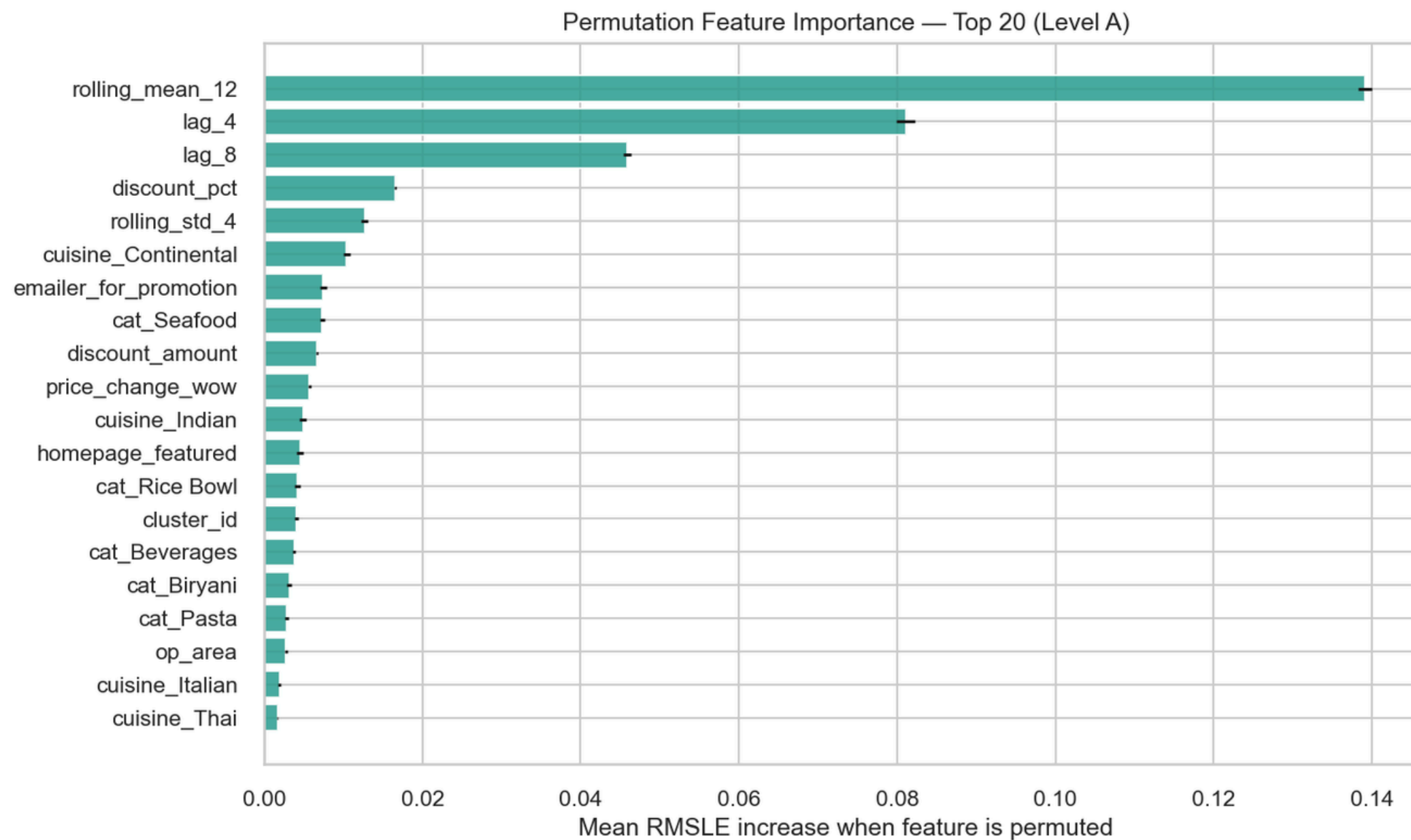
0.5190 ± 0.011

Tuning gain

marginal

PERMUTATION IMPORTANCE

RMSLE increase when each feature is shuffled — what the tuned model actually uses

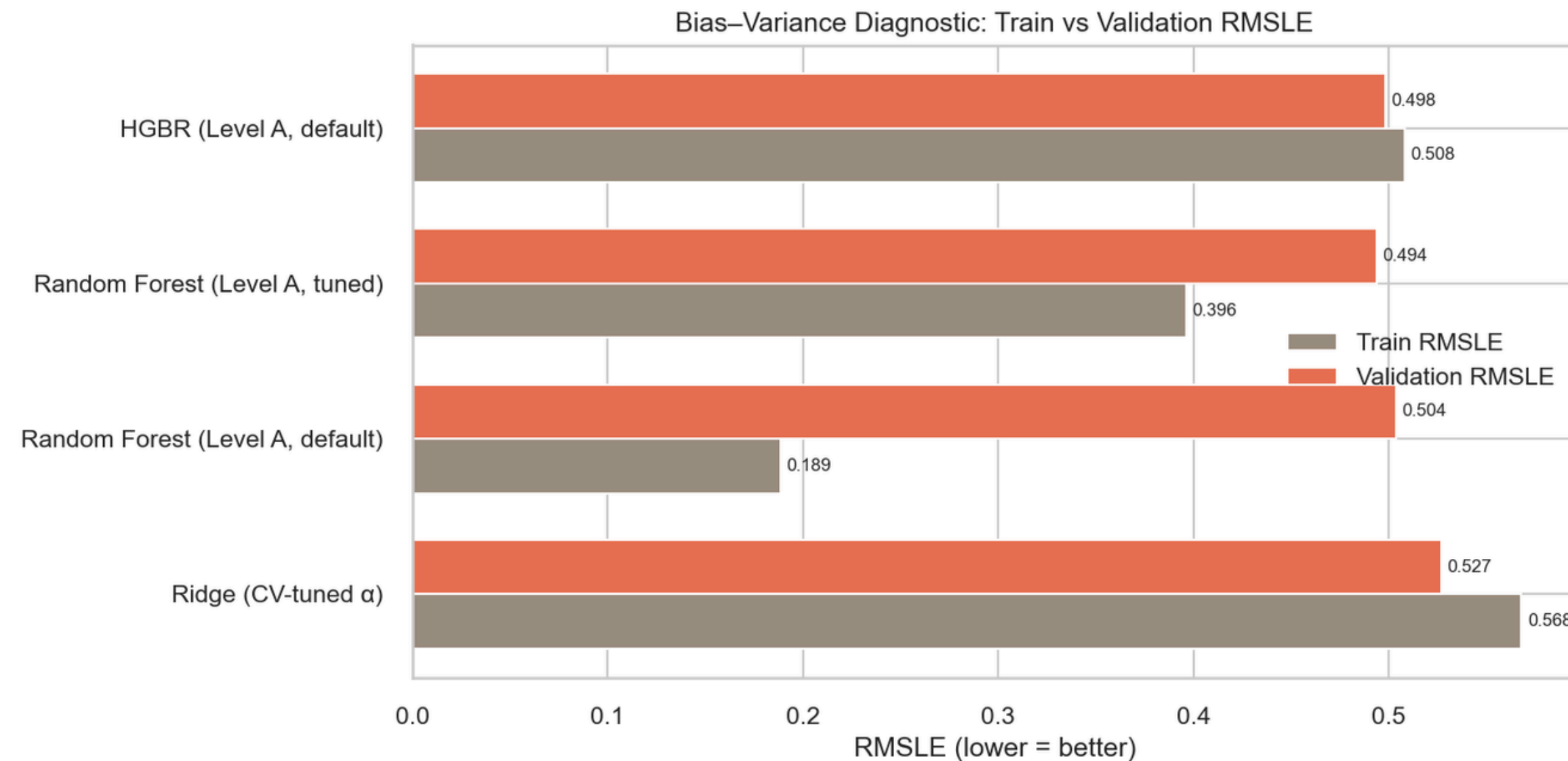


WHAT DRIVES THE MODEL

- **Temporal features dominate** as rolling_mean_12 (0.143), lag_4 (0.084), lag_8 (0.049) — the top three carry most predictive power.
- **rolling_mean_12 is ~1.7× the next** meaning every non-temporal predictor scores below 0.02.
- **Promotion & price are secondary** as discount_pct, emailer, homepage — they modulate demand around the temporal baseline.

BIAS-VARIANCE DIAGNOSTICS

Train vs validation RMSLE — where each model sits on the spectrum



READING THE GAPS

- **RF default** — **gap +167%** means severe overfitting (unbounded depth, leaf = 1).
- **RF tuned** — **gap +25%** means variance controlled; accuracy barely moves (0.498→0.508).
- **HGBR default** — **gap -2%** means near-ideal generalisation out of the box.
- **Ridge** — **negative gap** means controlled underfitting; the additive ceiling.

Tuning's job here is variance reduction, not accuracy — a third signal of the predictability ceiling.

CLUSTER-SEGMENTED MODELLING (LEVEL B)

One HGBR per cluster vs one global HGBR — hyperparameters fixed to isolate segmentation

LEVEL A — GLOBAL

0.4984

[0.4926, 0.5044]

LEVEL B — SEGMENTED

0.4985

[0.4925, 0.5045]

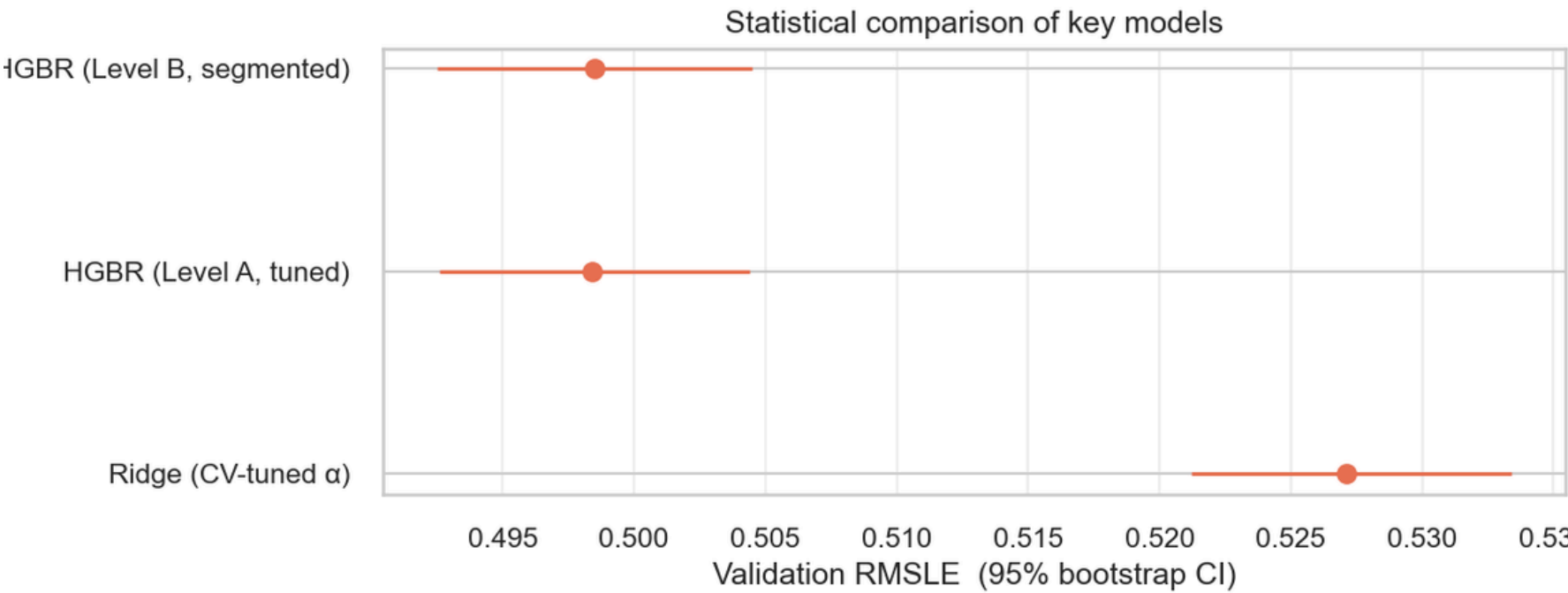
DIFFERENCE

-0.0001

CIs overlap → not significant

PER-CLUSTER Δ (Level B – Level A)

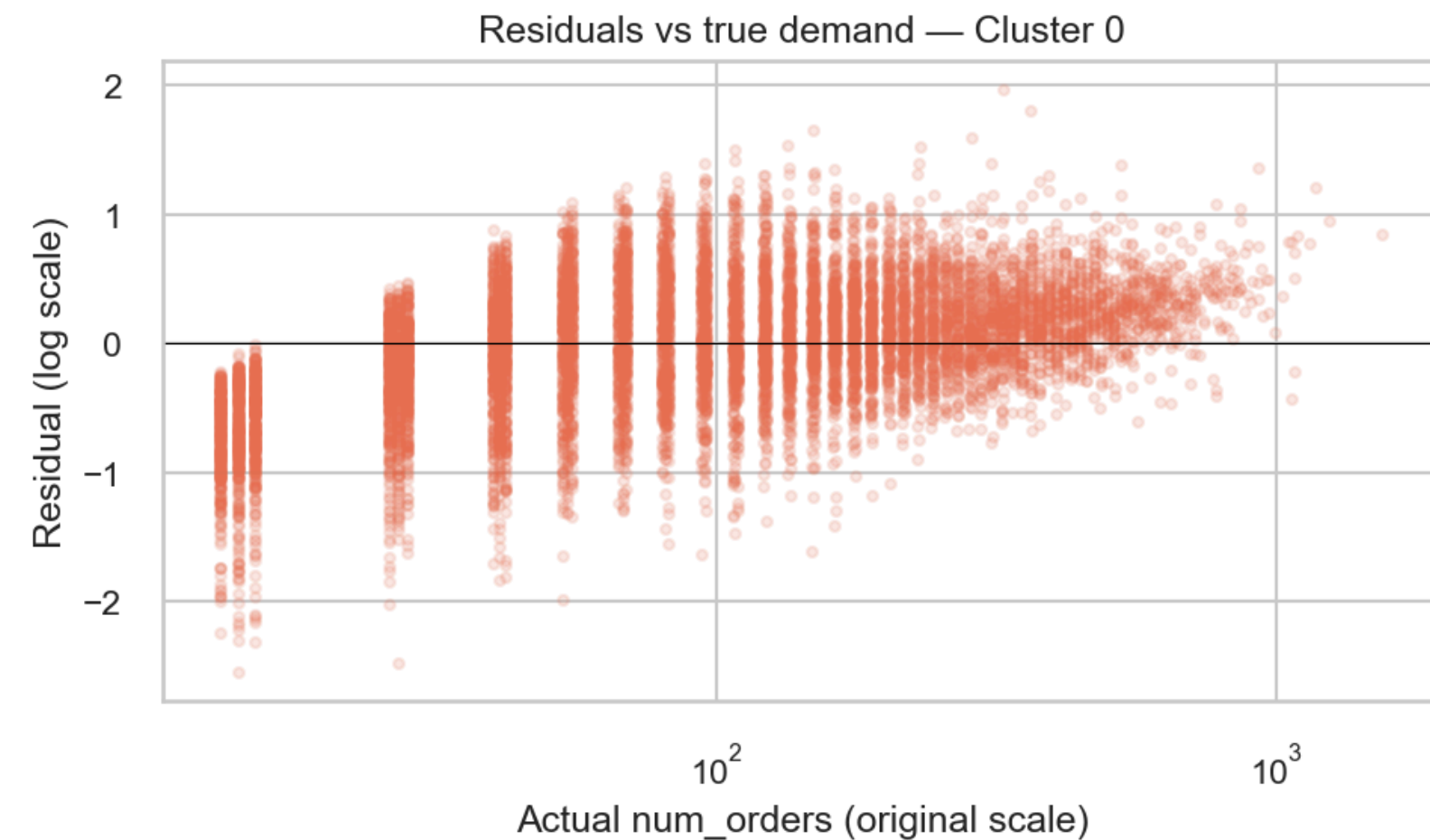
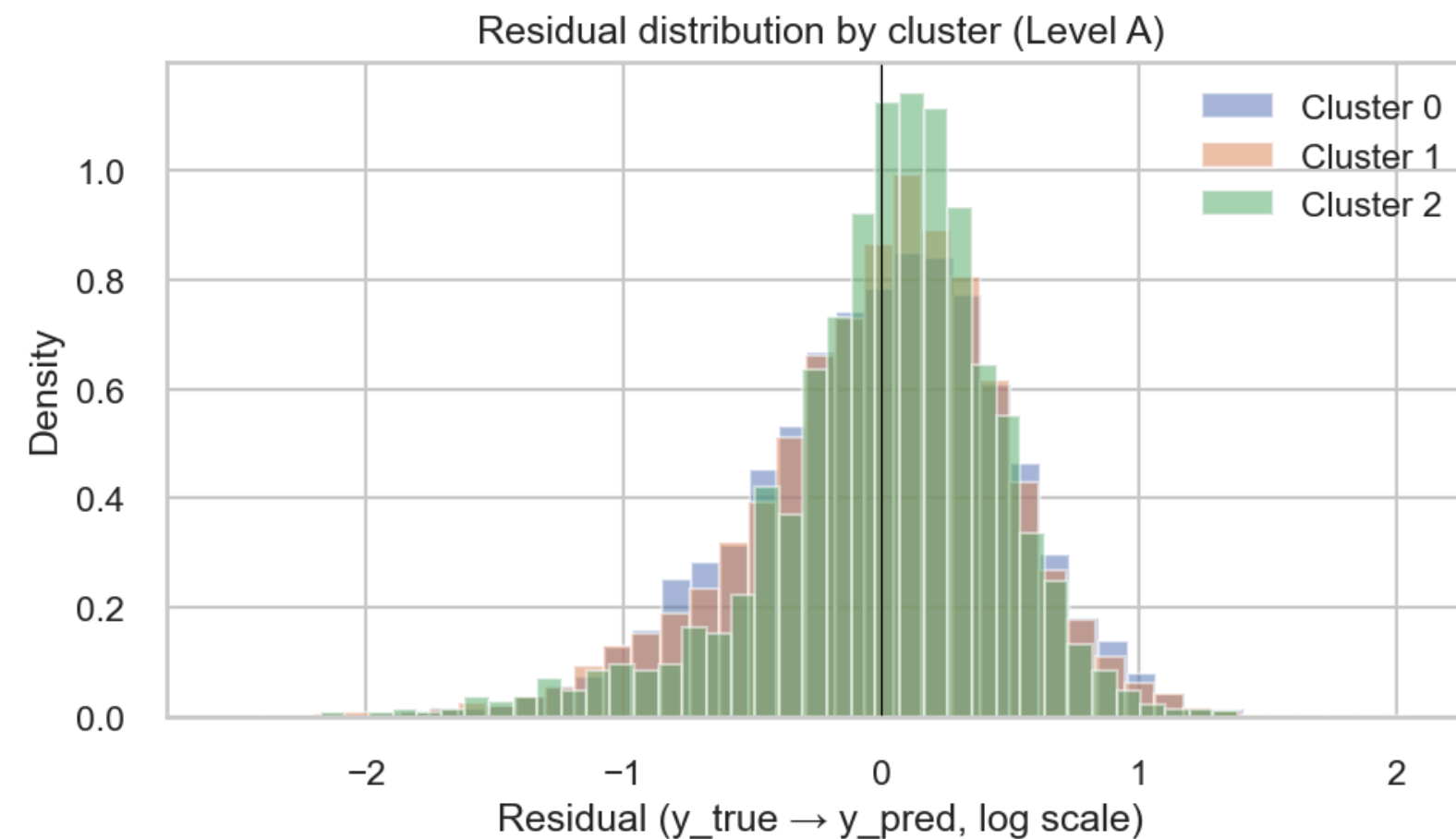
CLUSTER	PROFILE	Δ RMSLE
0 · Type B	high volume, low volatility	-0.0006
1 · Type C	low volume, highest volatility	+0.0006
2 · Type A	mid volume, price-sensitive	-0.0007



The hypothesised partial-pooling bias does not appear. **The global HGBR already holds cluster_id and the temporal features (lag_4/8, rolling_mean_12) that implicitly encode each center’s demand regime — so physical segmentation adds essentially nothing.**

DIAGNOSTICS: WHY CLUSTER 1 IS HARDEST

Residual analysis: is the hardest cluster biased, or just intrinsically noisy?



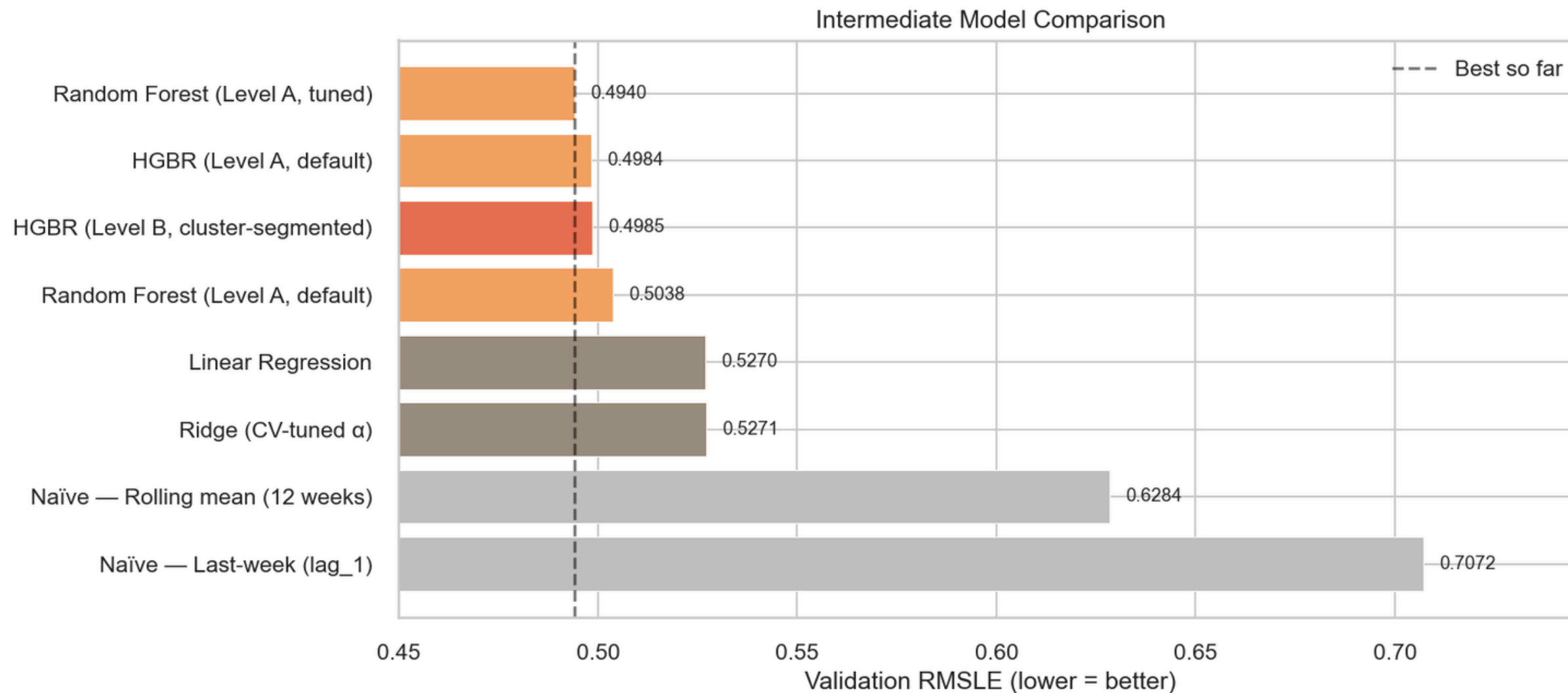
Cluster	Mean	CV	RMSLE
0	291	1.22	0.461
2	222	1.18	0.499
1  noisiest	179	1.12	0.518

IT'S A PREDICTABILITY CEILING, NOT A GAP

- **Unbiased everywhere** — mean residuals ≈ 0 in all clusters (0.018, -0.020, -0.021).
- **Cluster 1 = highest noise-to-signal** — largest fluctuations relative to its low demand — exactly what RMSLE penalises.
- **No structure in residuals** — fan-shaped scatter, not systematic over/under-prediction — little signal left to exploit.

INTERMEDIATE SUMMARY

Three layers of separation summarise the whole pipeline so far



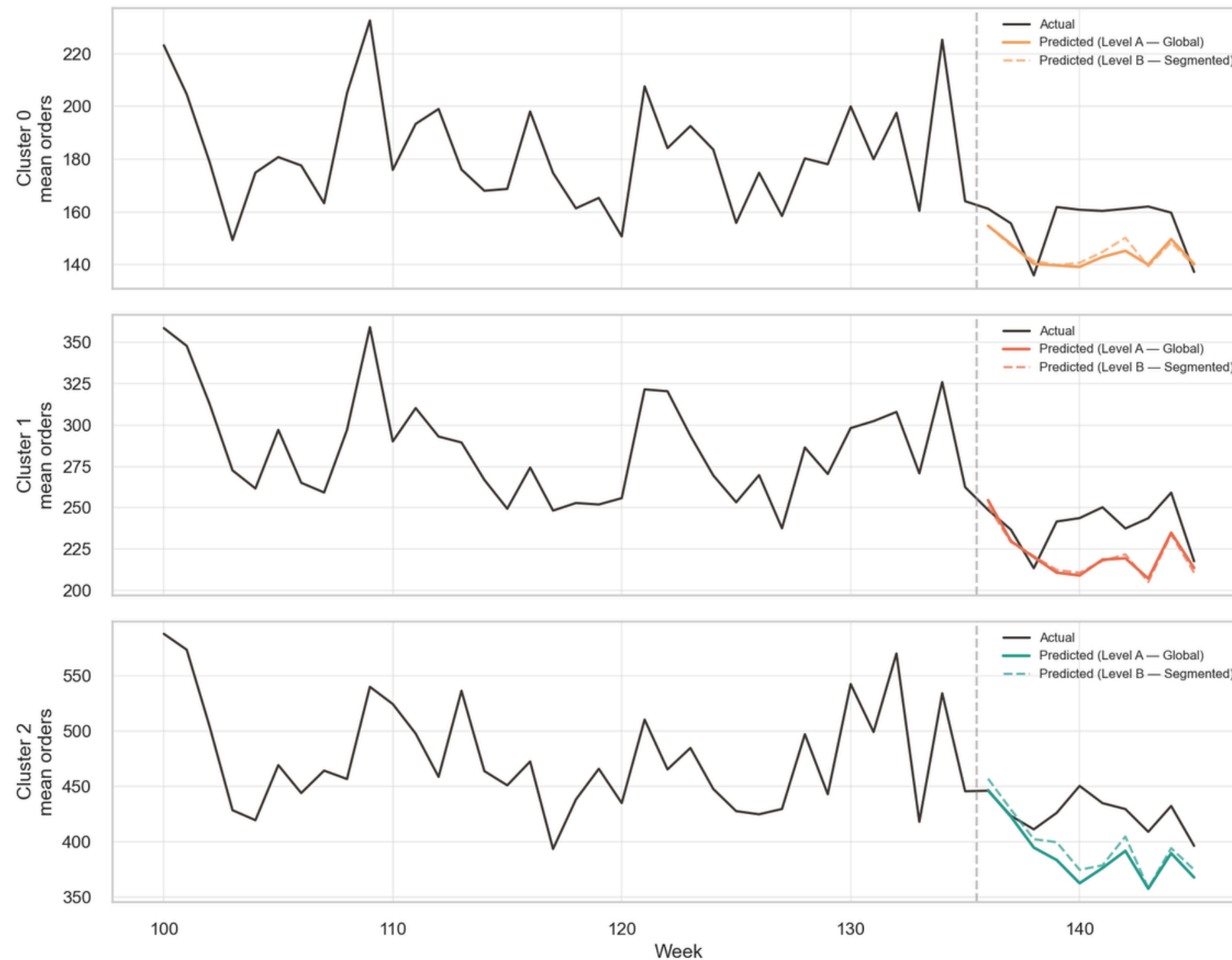
KEY OBSERVATIONS

- **Naïve floor (0.7072 / 0.6284)** — every learned model clears it comfortably.
- **Linear plateau (~0.527)** LR, Ridge, cluster-Ridge all cluster here — the additive ceiling.
- **Non-linear plateau (~0.494–0.508)** all 4 ensemble configs within 0.0125 RMSLE — not statistically separable.
- **Hierarchy: Naïve < Linear < Non-linear** best model is 20.9% better than the floor; segmentation effect stays tiny.

ACTUAL vs PREDICTED BY CLUSTER

Level A (solid) vs Level B (dashed) over the validation window, weeks 136–145

B.5 Actual vs Predicted Demand — Level A vs Level B, by Cluster



WHAT THE CURVES SHOW

- **A and B are indistinguishable** - solid and dashed lines overlap so closely they read as one trajectory in every cluster.
- **Level captured** - models track each archetype's typical demand band well.
- **Shocks are not anticipated** - week-to-week deviations remain — the residual is irreducible uncertainty, not missing complexity.
- **Confirms the null result** - cluster-aware modelling does not change the forecast under the full feature regime.

CLUSTER-RIDGE vs GLOBAL HGBR

Asymmetric Format 4 test: can structural specialisation replace model capacity?

3× Cluster-Ridge

0.5262

~7% worse · bootstrap CIs do NOT overlap

Global HGBR

0.4984

PER-CLUSTER Δ (HGBR $\rightarrow$ Ridge)

Cluster	Ridge	HGBR	Δ
0	0.5356	0.5119	-0.024
1	0.5290	0.4963	-0.033
2	0.4841	0.4577	-0.026

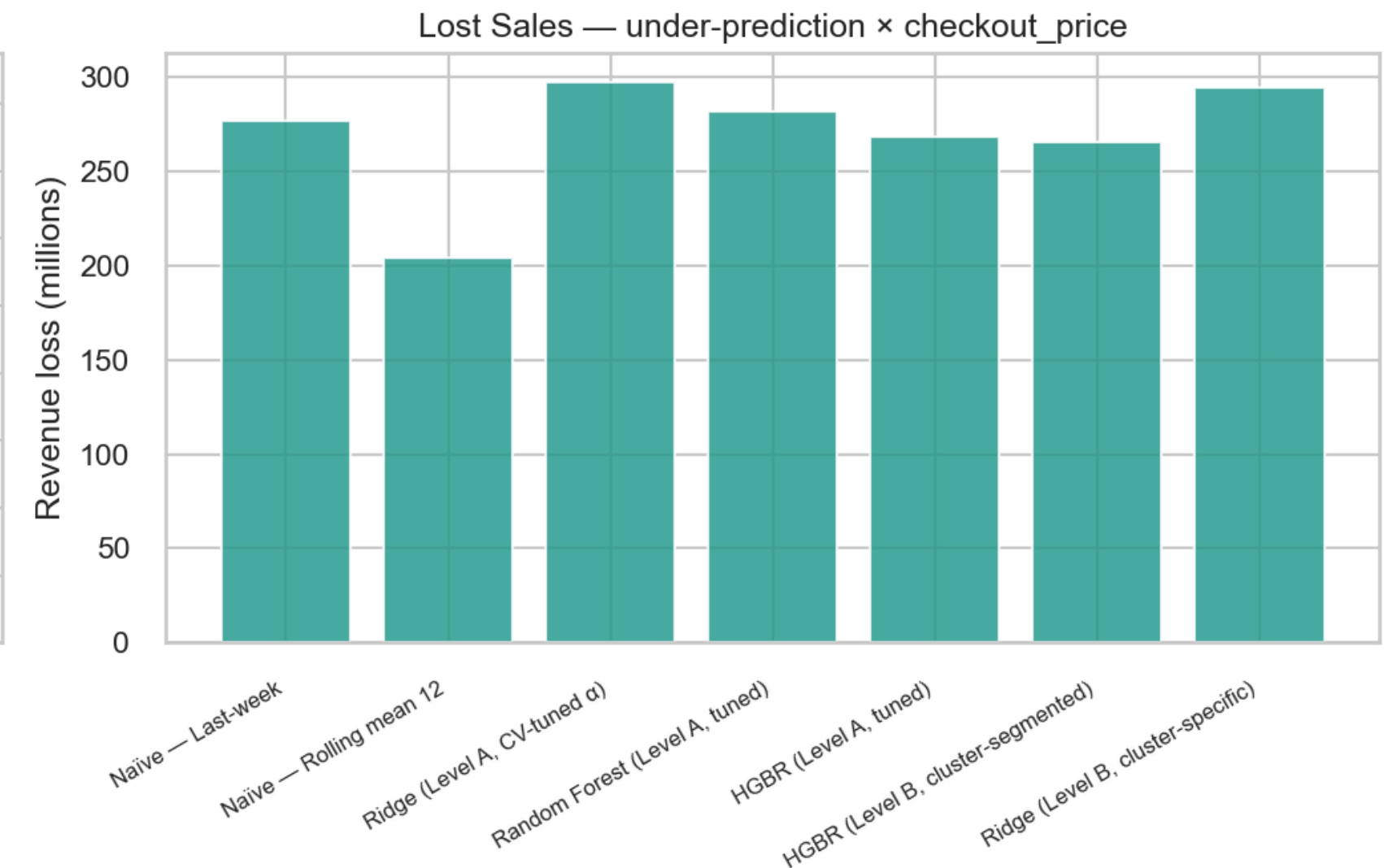
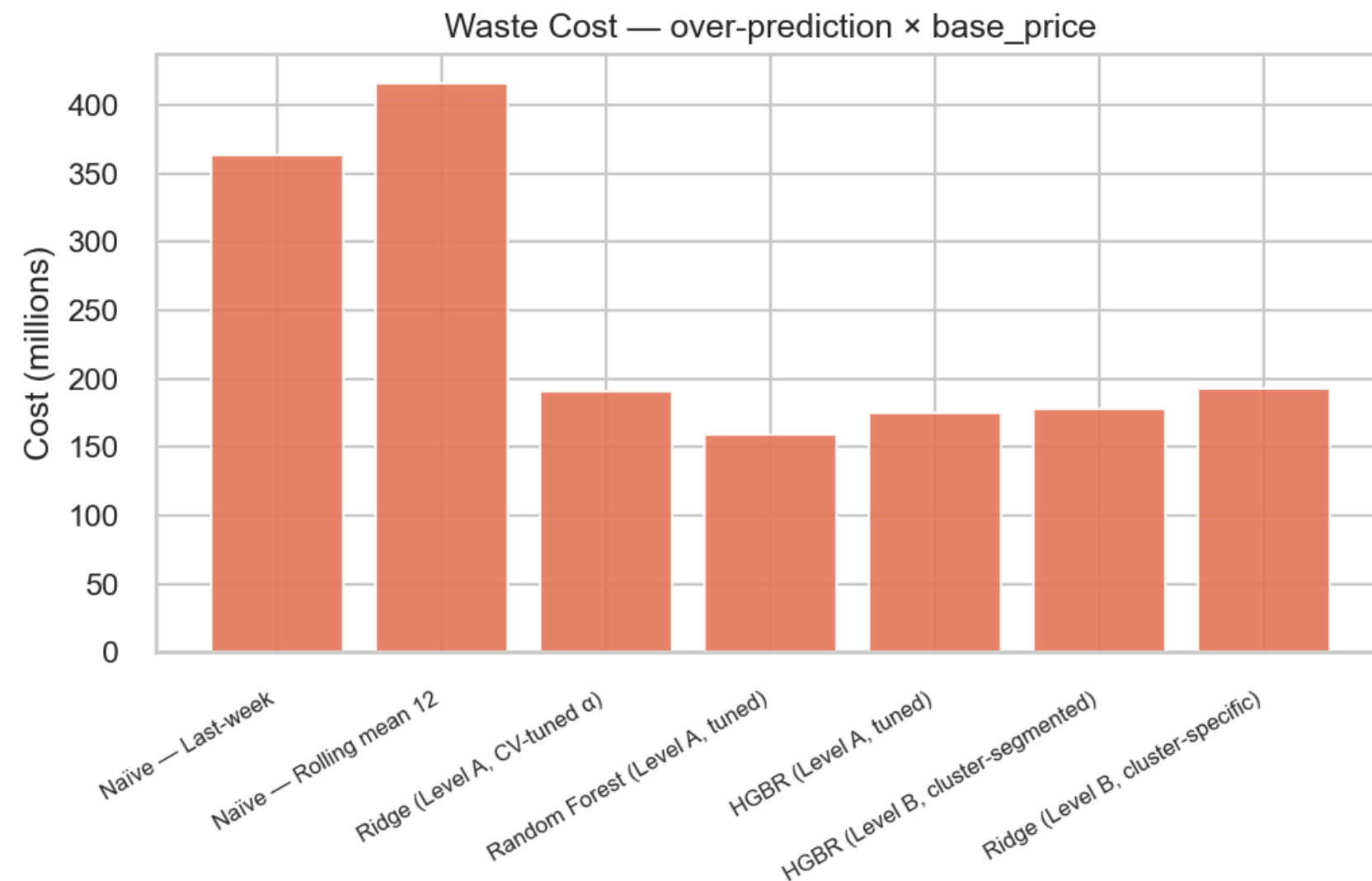
SPECIALISATION $\neq$ CAPACITY

- **Ridge loses in every cluster** - the degradation is broad, not confined to one problematic segment.
- **The win is non-linear interactions** - price $\times$ promotion $\times$ temporal structure that no additive model can represent, however finely segmented.
- **Confirms the through-line** - $A \approx B$: gap stable without the dominant feature and Ridge < HGBR.
- **Model class is the lever** - the decisive jump is linear $\rightarrow$ non-linear, not partitioning the data.

BUSINESS IMPACT LAYER

Translating forecast error into euros — over-prediction → waste, under-prediction → lost sales

Business Impact by Model — Validation Set

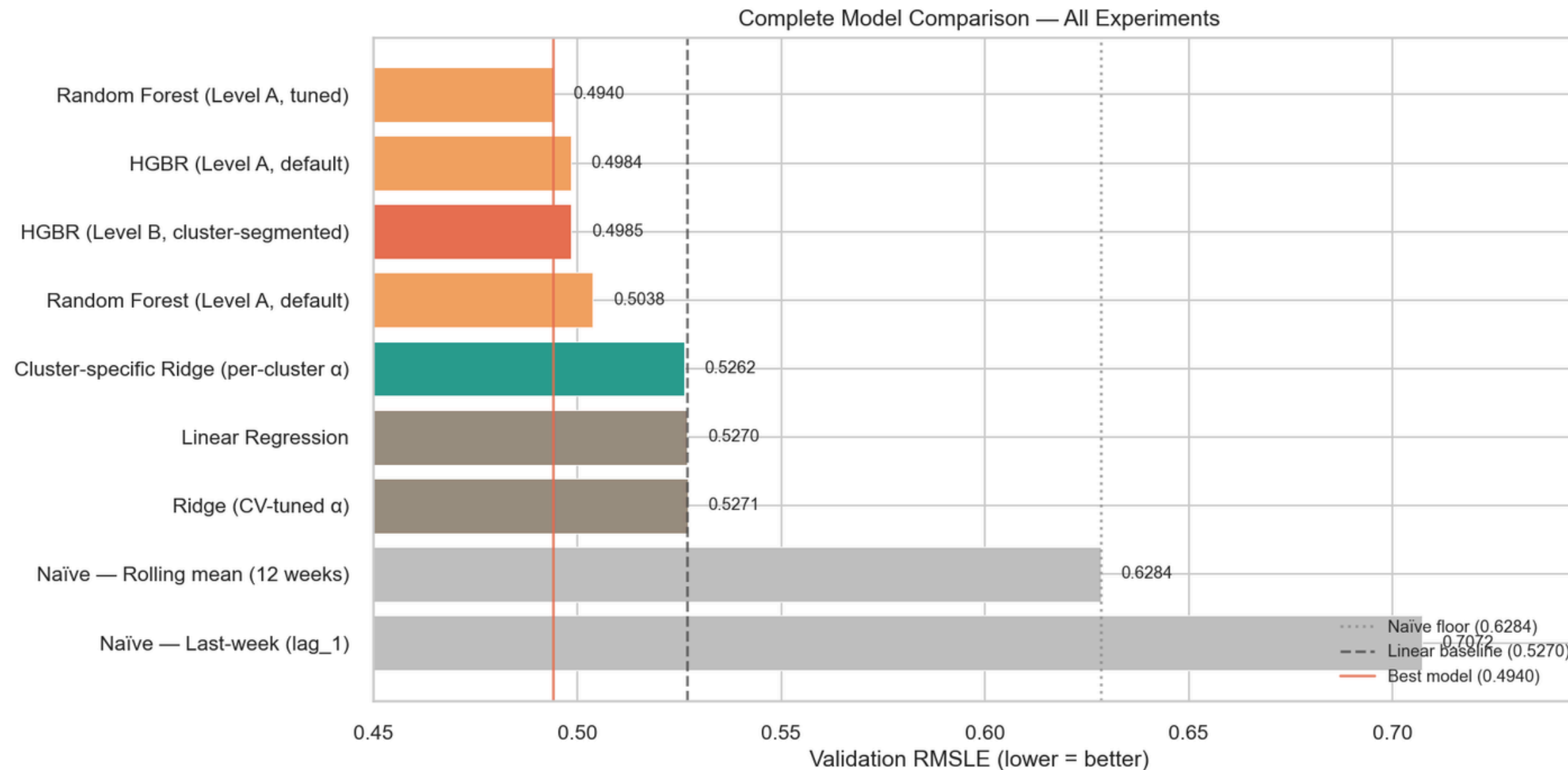


COST SAVINGS

- **Level B costs ~1.8% more waste**
segmentation gives no operational edge either.
- **HGBR A / B → 30.7%**
all learned models collapse into one cost band.
- **Random Forest A → 31.1%** best operational result vs the worst naïve baseline (€640M).
- **Consistent shortage bias** - all good models under-predict slightly — lost sales exceed waste.

FINAL SUMMARY — ALL EXPERIMENTS

The complete model ladder, from naïve floor to tuned ensembles



THE ANSWER

- **RQ verdict: not distinguishable** as global 0.4955 vs segmented 0.4971 — inside the bootstrap noise band.
- **Pipeline justified: –20.9%** as linear adds ~15%, non-linear a further ~7% — each layer earns its place.
- **Model class is the dominant lever** as linear→non-linear is the only significant jump; segmentation never leaves the plateau.
- **Format 4 confirms it** cluster-Ridge loses to global HGBR by ~7% with non-overlapping CIs.

CONCLUSION

**Segmentation does not reduce error —
model class does.**

PIPELINE JUSTIFIED

Best model cuts RMSLE 20.9% vs the naïve floor — gains accumulate at every layer.

FORMAT 4 CONFIRMS

Cluster-Ridge loses to global HGBR by ~7% (non-overlapping CIs). Specialisation ≠ capacity.

MODEL CLASS WINS

Linear → non-linear is the only significant jump. The 4 non-linear configs sit within 0.0125 RMSLE.

WHERE IT HELPS

Only Cluster 1 (highest volatility) gains from a local model once the dominant feature is removed.

Contribution: clustering is the wrong lever for accuracy on this dataset — it earns its place as an interpretability and diagnostics layer.



Thank you

Teodora Atanasova & Martina Puglisi